

MUHAMMAD ZEESHAN

Mechatronics Engineering | Robotics & Automation | Embedded Systems

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PROFILE

Driven 3rd-year Mechatronics Engineering undergraduate at MUET Jamshoro (CGPA 3.18/4.00) with a strong cross-disciplinary foundation in robotics, embedded systems, automation, and AI. Certified by Northwestern University in Modern Robotics and equipped with hands-on project experience, professional safety training, and IEEE-recognised learning. Seeking an internship to contribute to cutting-edge engineering projects and translate academic expertise into industry impact.

EDUCATION

Bachelor of Engineering — Mechatronics

Aug 2024 – Present

Mehran University of Engineering & Technology (MUET), Jamshoro · CGPA: 3.18 / 4.00 · 3rd Year

- Relevant Courses: Engineering Mechanics, Electrical Circuits & Electronics, Control Systems, Microcontrollers, CAD & Manufacturing Processes

HSSC — Pre-Engineering

Jul 2022 – May 2024

Government Boys High Secondary School, Hayat Pitafi · Marks: 813 / 1100

SSC — Science

Aug 2020 – May 2022

Government Boys High Secondary School, Hayat Pitafi · Marks: 886 / 1100

CERTIFICATIONS & SPECIALIZATIONS

Modern Robotics: Mechanics, Planning, and Control — Specialization

Northwestern University

Coursera · Comprehensive specialization covering robot kinematics, dynamics, motion planning, and control theory.

AI-Aided Drive System Design, Control & Maintenance for More-Electric Aircraft

IEEE

Instructor: Yuan Gao · Covered AI applications in electric aircraft propulsion systems and condition monitoring.

Machine Learning Enabled Design Automation & Optimization for Electric Transportation Power Systems

IEEE

Applied ML techniques for automated design and performance optimization in electric vehicle power systems.

Artificial Intelligence (Machine Learning, Deep Learning, Communication)

NAVTTTC

Prime Minister's Digital Learning Programme — Batch 1 · Completed structured AI/ML curriculum with practical modules in deep learning and data communication.

PROFESSIONAL TRAINING

Hazard Identification and Risk Assessment (HIRA)

2025

ISO & ASCE-MUET Student Chapter · Trainer: Engr. Mahmood Ali Arain, CEO LEAD HSE · Duration: 5 Days

- Acquired structured methodology for identifying workplace hazards and evaluating risk severity in industrial environments.

Project-Based Arduino Training

Oct 2024

Self-Initiated / Workshop-Based

- Designed and built circuits from basic LED control to sensor-actuator integration using Arduino IDE.

PROJECTS

Robotics Fleet Management System

MUET

- Designed a coordinated multi-robot management framework covering task allocation, path planning, and collision avoidance for autonomous robotic fleets.
- Integrated sensor feedback and control logic to enable real-time status monitoring and dynamic re-routing.

Water Turbidity Measuring System

2024

- Developed a real-time water quality monitoring device using a turbidity sensor and Arduino microcontroller to quantify suspended particle concentration.
- Calibrated across multiple NTU levels with serial data logging for continuous field deployment.

TECHNICAL SKILLS

Robotics & Automation: Modern Robotics (Northwestern), Fleet Management, PLC Fundamentals, Robotics Basics, Soft Robotics (Research Interest)

Embedded Systems: Arduino, Microcontroller Programming, Sensor Integration, PCB Design (Proteus / Multisim)

AI & Machine Learning: Machine Learning, Deep Learning, ML for Design Automation, AI for Electric Systems (IEEE)

CAD & Design: SolidWorks, SolidWorks Simulation, AutoCAD, Engineering Drawing, MATLAB & Simulink

Programming: C, C++ (Basic), Python, Arduino IDE

Other: Basic Circuit Analysis, Workshop Practices, MS Office (Excel, Word, PowerPoint)

HONORS & ACHIEVEMENTS

- Completed Northwestern University Specialization in Modern Robotics — Mechanics, Planning & Control
- Earned IEEE educational certificates in AI for Electric Aircraft and ML for Electric Transportation Systems
- Completed NAVTTC PM Digital Learning Programme — Artificial Intelligence (Batch 1)
- Completed HIRA Safety Training — ISO & ASCE-MUET Student Chapter (2025)
- Completed Project-Based Arduino Microcontroller Training (2024)

ATTRIBUTES & INTERESTS

- Fast learner and highly adaptable across technical and professional environments
- Strong problem-solving mindset with passion for robotics and autonomous systems research
- Collaborative team player with clear technical communication skills
- Active self-directed learner — consistently working on personal engineering projects
- Interests: Soft Robotics Research, Table Tennis, Cricket, Reading

REFERENCES

Dr. Jawaid Daudpoto

Chairman, Mechatronics Engineering — MUET

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Raheel Nizamani

Assistant Professor — MUET

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